

J.D. Peiffer

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Education

Northwestern University, Shirley Ryan AbilityLab

Chicago, IL

MS, PhD in Biomedical Engineering

Sept 2022 – present

- GPA: 3.81 | NSF Graduate Research Fellowship (2022); NIH T32 Training Grant (2024)

Washington University in St. Louis

St. Louis, MO

Rotation Student in Biomedical Engineering

Aug 2021 – July 2022

University of Missouri

Columbia, MO

BS in Biomedical Engineering

Aug 2017 – May 2021

- GPA: 3.92 | Emphasis: Bioinformatics | Minor: Mathematics
- Honors Scholar; Dean's List (all semesters)

Experience

PhD Researcher

Chicago, IL

Center for Bionic Medicine, Shirley Ryan AbilityLab

Sept 2022 – present

- Led first-author research in computer vision, sensor fusion, and biomechanical modeling for clinical AI, taking models from concept through validated clinical deployment; 15+ publications, >170 citations in <4 years.
- Developed and released the Portable Biomechanics Laboratory — the first clinically validated single-camera smartphone gait analysis system, achieving <3° joint-angle error and ICC > 0.9 for derived gait biomarkers; first-author paper in npj Digital Medicine (Nature).
- Deployed PBL to 3 external clinical sites (Shriners Children's, Lake Lucerne Institute), each of which published independent validation studies; demonstrated derived gait metrics detect surgical response in cervical myelopathy more sensitively than standard patient-reported outcomes.
- Designed a neural-network fusion of uncalibrated IMU and monocular smartphone video, cutting knee angle error 26% while maintaining accuracy through visual occlusions.
- Co-authored B3D, a feed-forward biomechanical prediction head on a frozen SAM-3D-Body foundation model, distilled label-free via differentiable MuJoCo physics; accepted to ECCV 2026 workshop.
- Built multi-camera calibration and SAM-based pseudo-labeling pipelines, generating pseudo-ground-truth annotations to train and benchmark monocular scene reconstruction models; curated a 10,000-video clinical movement dataset for scene understanding model development.
- Trained a contrastive learning + causal masking transformer on unannotated clinical gait data; demonstrated learned embeddings classify neurological diagnosis and track patient response to inpatient therapy.

Research Scientist Intern

Redmond, WA

Glove Sensing and Tracking Team, Meta Reality Labs

June 2025 – Feb 2026

- Implemented a real-time hand pose estimation and inverse-kinematics pipeline achieving <5 mm fingertip tracking error at 60 Hz with occlusion robustness across a 22-DoF hand model.
- Trained a CNN+Transformer model on fiber-optic sensor data for direct hand pose regression, benchmarking learned pose estimation against an inverse-kinematics baseline.
- Collected a dexterous-manipulation benchmarking dataset with motion-capture ground truth and authored production evaluation pipeline; first-author paper accepted to IROS 2026.

Fulbright-Mitacs Globalink Intern

Alberta, CA

Neuromuscular Control & Biomechanics Laboratory, University of Alberta

May 2020 – Aug 2020

- Computationally modeled seated human trunk biomechanics under Functional Electrical Stimulation (FES) and designed LQR and fuzzy inference (FIS) feedback controllers; validated closed-loop stability in simulation.

Undergraduate Researcher

Neural Engineering Laboratory, University of Missouri

Columbia, MO

Aug 2020 – May 2021

- Developed and optimized a closed-loop gamma oscillation detection and modulation algorithm achieving <2 ms latency on embedded hardware; open-sourced on [GitHub](#).

Undergraduate Researcher

Pulmonary Imaging Research Laboratory, University of Missouri

Columbia, MO

Aug 2018 – May 2021

- Built a semi-automated image segmentation, registration, and quantification pipeline enabling large-cohort comparison of 129Xe MRI vs. 99mTc SPECT; first-author publication in Academic Radiology (2024).
- Won Audience Choice Award for oral presentation at International Workshop on Pulmonary Functional Imaging (2019).

Selected Talks

- **Harnessing movement as a biomarker through deployable motion capture**, Invited talk to Eli Lilly & Human Longevity Lab (2026)
- **Deployable motion capture for in-clinic and in-the-wild biomechanical measurement**, Computerized Motion Analysis Laboratory Visiting Professor Program, Chicago, IL (2026)
- **Leveraging AI for Real-World Biomechanical Reconstruction from a Single Moving Camera**, European Society of Biomechanics, Zurich, Switzerland (2025)
- **Differentiable Biomechanics for Markerless Motion Capture in Upper Limb Stroke Rehabilitation**, International Consortium for Rehabilitation Robotics, Chicago, IL (2025)
- **Accurate Lower Body Kinematics Using a Handheld Smartphone**, American Society of Biomechanics Annual Meeting, Madison, WI (2024)

Selected Honors

- **NSF Graduate Research Fellowship**, National Science Foundation (2022)
- **Fulbright-Mitacs Globalink Internship** (2020)
- NIH T32 Training Grant (2024)
- 2nd Place Best Student Paper, ICORR (2025)
- Barry M. Goldwater Scholarship Campus Nomination (2019)
- Audience Choice Award, International Workshop on Pulmonary Functional Imaging (2019)
- 2 x Boston Marathon Qualifier (2024, 2025)
- Cherng Summer Scholar, University of Missouri (2019)

Skills

Languages: Python, C++, R

ML Frameworks: JAX (Jaxlie), PyTorch (Pytorch3D), TensorFlow, MuJoCo/MJX, OpenCV

Infrastructure: Docker, Kubernetes, Git, SQL, Datajoint, W&B

Research Areas: Computer Vision, Sensor Fusion, Biomechanics, Wearables, Self-Supervised Learning, Generative Models

Publications

Biomechanics-aware Multi-view Markerless Motion Capture of Dexterous Hand Movements

July 2026

Pouyan Firouzabadi, *J.D. Peiffer*, Kunal Shah, Anton Sobinov, Lee E. Miller, R. James Cotton, Wendy M. Murray [10.48550/arXiv.2607.02796](#) (arXiv)

- Biomechanical 3D Body: Self-Supervised Distillation of Biomechanical Pose from a 3D Body Foundation Model** Aug 2026
R. James Cotton, *J.D. Peiffer*, Lucinda A. Williamson, John Leske, Georgios Pavlakos
[10.48550/arXiv.2608.29928](https://arxiv.org/abs/10.48550/arXiv.2608.29928) (ECCV 2026 Workshop MoCha)
- Fiber Optic Sensing Glove for High Performance Dexterous Manipulation Capture** Aug 2026
J.D. Peiffer, Taylor Niehues, Li Guan, Ziyi Kou, Ergys Ristani
[10.48550/arXiv.2608.24572](https://arxiv.org/abs/10.48550/arXiv.2608.24572) (IROS (Accepted))
- Portable biomechanics laboratory enables clinically accessible movement analysis from a handheld smartphone** June 2026
J.D. Peiffer, Kunal Shah, Irina Djuraskovic, Shawana Anarwala, Kayan Abdou, Rujvee Patel, Prakash Jayabalan, Brenton Pennicooke, R. James Cotton
[10.1038/s41746-026-02825-w](https://doi.org/10.1038/s41746-026-02825-w) (npj Digital Medicine)
- Markerless Motion Capture for Biomechanical Whole-Body Kinematic Estimation in Infants** May 2026
Divya Joshi, *J.D. Peiffer*, Colleen Peyton, R. James Cotton
[10.48550/arXiv.2605.17120](https://arxiv.org/abs/10.48550/arXiv.2605.17120) (EMBC (Accepted))
- Monocular Markerless Motion Capture Enables Quantitative Assessment of Upper Extremity Reachable Workspace** May 2026
Seth Donahue, *J.D. Peiffer*, R. Tyler Richardson, Yishan Zhong, Shaun Q. Y. Tan, Benoit L. Marteau, Stephanie A. Russo, May D. Wang, R. James Cotton, Ross Chafetz
[10.3390/s26113421](https://doi.org/10.3390/s26113421) (Sensors, 26(11), 3421)
- Differentiable Biomechanics for Markerless Motion Capture in Upper Limb Stroke Rehabilitation: A Comparison With Optical Motion Capture** Feb 2026
Tim Unger, Arash Sal Moslehian, *J.D. Peiffer*, Johann Ullrich, Roger Gassert, Olivier Lambercy, R. James Cotton, Chris Awai Easthope
[10.1109/TMRB.2025.3605962](https://doi.org/10.1109/TMRB.2025.3605962) (IEEE Transactions on Medical Robotics and Bionics | 2nd Place Best Student Paper, ICORR 2025)
- Synergy-based functional electrical stimulation for gait rehabilitation in chronic stroke: a pilot study** Aug 2026
Jackson Taylor Levine, Xin Yu, Rebecca Muñoz, Alaina Fiorenza, Irina Djuraskovic, *J.D. Peiffer*, Emilia Ambrosini, Simona Ferrante, Elliot J. Roth, Jozsef Laczko, R. James Cotton, Alessandra Pedrocchi, Jose L. Pons
[10.3389/fresc.2026.1847719](https://doi.org/10.3389/fresc.2026.1847719) (Frontiers in Rehabilitation Sciences)
- Fusing Uncalibrated IMUs and Handheld Smartphone Video to Reconstruct Knee Kinematics** Sept 2024
J.D. Peiffer, Kunal Shah, Shawana Anarwala, Kayan Abdou, R. James Cotton
[10.1109/BioRob60516.2024.10719724](https://doi.org/10.1109/BioRob60516.2024.10719724) (BioRob 2024)
- Biomechanical Arm and Hand Tracking with Multiview Markerless Motion Capture** Sept 2024
Pouyan Firouzabadi, Wendy Murray, Anton R Sobinov, *J.D. Peiffer*, Kunal Shah, Lee E Miller, R. James Cotton
[10.1109/BioRob60516.2024.10719940](https://doi.org/10.1109/BioRob60516.2024.10719940) (BioRob 2024)

- Hyperpolarized 129Xe MRI, 99mTc scintigraphy, and SPECT in lung ventilation imaging: a quantitative comparison** Apr 2024
J.D. Peiffer, Talissa Altes, Iulian C. Ruset, F. W. Hersman, John P. Mugler, Craig H. Meyer, Jamie Mata, Kun Qing, Robert Thomen
[10.1016/j.acra.2023.10.038](https://doi.org/10.1016/j.acra.2023.10.038) (Academic Radiology)
- Optimizing Trajectories and Inverse Kinematics for Biomechanical Analysis of Markerless Motion Capture Data** Sept 2023
 R. James Cotton, Allison DeLillo, Anthony Cimorelli, Kunal Shah, *J.D. Peiffer*, Shawana Anarwala, Kayan Abdou, Tasos Karakostas
[10.1109/ICORR58425.2023.10304683](https://doi.org/10.1109/ICORR58425.2023.10304683) (ICORR 2023)
- Self-Supervised Learning of Gait-Based Biomarkers** Sept 2023
 R. James Cotton, *J.D. Peiffer*, Kunal Shah, Allison DeLillo, Anthony Cimorelli, Shawana Anarwala, Kayan Abdou, Tasos Karakostas
[10.1007/978-3-031-46005-0_24](https://doi.org/10.1007/978-3-031-46005-0_24) (Predictive Intelligence in Medicine (Springer))
- Enhanced selectivity of transcutaneous spinal cord stimulation by multielectrode configuration** Aug 2023
 Noah Bryson, Lorenzo Lombardi, Rachel Hawthorn, Jie Fei, Rodolfo Keeseey, *J.D. Peiffer*, Ismael Seáñez
[10.1088/1741-2552/ace552](https://doi.org/10.1088/1741-2552/ace552) (Journal of Neural Engineering)

Teaching

- Teaching Assistant — Biomedical Applications of Machine Learning** *Chicago, IL*
Northwestern University Mar 2025 – June 2025
 - Instructed graduate students in Python-based ML including data preprocessing, regression, tree-based methods, DNNs, and CNNs.
- Teaching Assistant — Advanced Hydroinformatics** *Columbia, MO*
University of Missouri Jan 2020 – May 2021
 - Created curriculum introducing graduate students to scientific data manipulation and parallel computation job submission on HPC clusters.
- Ski Instructor** *Dec 2018 – present*
Vail Resorts

Service

- Speaker** *Apr 2023 – Apr 2025*
National Biomechanics Day
 Developed interactive sessions for high school students on biomechanics, prosthetics, computer vision, and AI for medical image analysis.
- Peer Reviewer** *Apr 2022 – present*
BioRob, IROS, iScience, CMBBE — Imaging & Visualization
- Site Leader** *Nov 2018 – Aug 2020*
Mizzou Alternative Breaks
 - Organized a week-long adaptive sports service trip (Ski Apache, NM) for 12 participants; assisted in instruction of students with visual, cognitive, and physical disabilities.